

⚠ This backtest uses only data available at each test date – no lookahead bias. The engine runs exactly as in live trading, sliced to the test date. Expect 5-15 minutes for a full run (46 test dates × full analogue search).

Pattern window

6 months (126 days)

Run Validation Backtest

Phase 1+2 Direction

Panel 3 Sector Rotation

Tests Panel 3's method: given the top analogues at each date, does investing in the sector that historically led in those analogue periods outperform SPY? Only uses post-1998 ETF data (no proxy baskets). Test window: 2002-2022.

Pattern window

6 months (126 days)

Run Panel 3 Validation

47s elapsed

Complete in 47.8s

48 / 48

BEAT SPY RATE

43.8%

vs 51.2% random baseline

LIFT OVER RANDOM

-7.4pp

percentage point improvement

AVG RELATIVE RETURN

-1.0%

recommended sector vs SPY

AVG SECTOR RANK

5.3 / 9

1=best, 9=worst out of 9 sectors

GOT BEST SECTOR

18.8%

exactly matched top sector

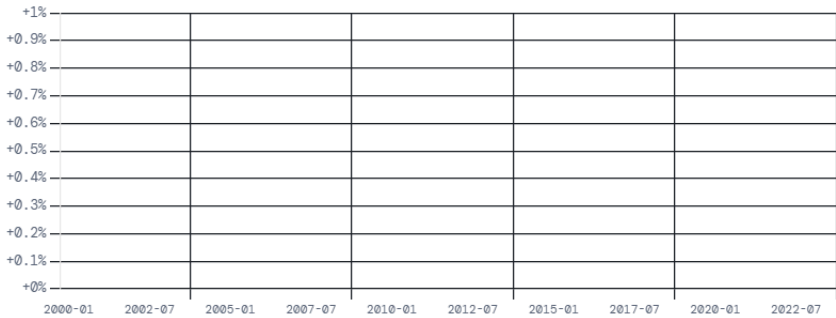
TEST DATES

48

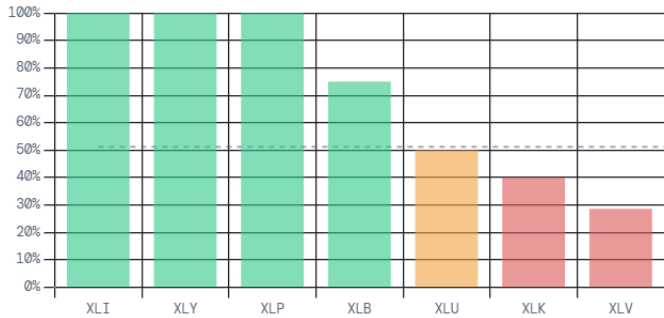
0 skipped (no ETF data)

Panel 3 underperforms random sector selection — the method needs improvement.
The recommended sector beat SPY only 43.8% of the time vs 51.2% baseline. The analogue-based sector prediction may be counterproductive.

Recommended sector actual relative return vs SPY over time



Beat SPY rate by recommended sector



ACCURACY BY RECOMMENDED SECTOR

Sector	Name	Times recommended	Beat SPY %	Avg relative return
XLI	Industrials	2	100%	+1.3%
XLY	Consumer Disc	1	100%	+5.1%
XLP	Cons Staples	1	100%	+2.1%
XLB	Materials	4	75%	+1.6%
XLU	Utilities	6	50%	+2.4%
XLK	Technology	20	40%	-1.0%
XLV	Healthcare	7	28.6%	-1.2%
XLE	Energy	6	16.7%	-7.0%
XLF	Financials	1	0%	-8.5%

Test date	Top recommendation	P3 score	Actual rel 12mo	Beat SPY?	Actual best	Sector rank	Macro
2000-01-01	XLK	25.8%	—	X No	XLE	6/9	Fed 5.5% YC 0.21%
2000-07-01	XLK	22.3%	—	X No	XLF	9/9	Fed 6.5% YC -0.35%
2001-01-01	XLK	36.1%	—	X No	XLV	7/9	Fed 6.0% YC 0.01%
2001-07-01	XLK	28.0%	—	X No	XLY	9/9	Fed 3.8% YC 1.17%
2002-01-01	XLI	6.3%	—	✓ Yes	XLB	8/9	Fed 1.7% YC 2.00%
2002-07-01	XLK	15.7%	—	✓ Yes	XLV	2/9	Fed 1.7% YC 1.97%
2003-01-01	XLK	9.8%	—	✓ Yes	XLU	3/9	Fed 1.2% YC 2.22%
2003-07-01	XLK	17.7%	—	✓ Yes	XLB	3/9	Fed 1.0% YC 2.26%
2004-01-01	XLK	18.7%	—	X No	XLE	9/9	Fed 1.0% YC 2.43%
2004-07-01	XLK	13.3%	—	X No	XLU	7/9	Fed 1.3% YC 1.93%
2005-01-01	XLK	35.3%	—	X No	XLE	7/9	Fed 2.3% YC 1.16%
2005-07-01	XLK	4.2%	—	✓ Yes	XLE	5/9	Fed 3.3% YC 0.30%
2006-01-01	XLK	13.5%	—	X No	XLE	8/9	Fed 4.3% YC -0.02%
2006-07-01	XLE	12.8%	—	X No	XLY	9/9	Fed 5.2% YC -0.01%
2007-01-01	XLF	3.8%	—	X No	XLE	9/9	Fed 5.3% YC -0.11%
2007-07-01	XLU	13.2%	—	✓ Yes	XLE	2/9	Fed 5.3% YC 0.16%
2008-01-01	XLK	5.9%	—	X No	XLE	8/9	Fed 3.9% YC 0.99%
2008-07-01	XLB	7.8%	—	X No	XLP	8/9	Fed 2.0% YC 1.38%
2009-01-01	XLV	13.1%	—	X No	XLK	4/9	Fed 0.1% YC 1.49%
2009-07-01	XLB	15.2%	—	✓ Yes	XLB	1/9	Fed 0.2% YC 2.50%
2010-01-01	XLE	16.7%	—	X No	XLY	8/9	Fed 0.1% YC 2.71%
2010-07-01	XLY	6.7%	—	✓ Yes	XLB	3/9	Fed 0.2% YC 2.33%
2011-01-01	XLI	6.3%	—	✓ Yes	XLV	5/9	Fed 0.2% YC 2.69%
2011-07-01	XLU	6.7%	—	✓ Yes	XLU	1/9	Fed 0.1% YC 2.72%
2012-01-01	XLE	5.7%	—	X No	XLK	9/9	Fed 0.1% YC 1.64%
2012-07-01	XLK	7.4%	—	X No	XLF	8/9	Fed 0.2% YC 1.34%
2013-01-01	XLE	8.0%	—	X No	XLV	6/9	Fed 0.1% YC 1.53%
2013-07-01	XLU	8.5%	—	X No	XLI	9/9	Fed 0.1% YC 2.16%
2014-01-01	XLU	5.5%	—	✓ Yes	XLU	1/9	Fed 0.1% YC 2.66%
2014-07-01	XLV	4.3%	—	✓ Yes	XLV	1/9	Fed 0.1% YC 2.11%
2015-01-01	XLE	7.5%	—	X No	XLV	8/9	Fed 0.1% YC 1.50%
2015-07-01	XLV	4.3%	—	X No	XLU	7/9	Fed 0.1% YC 1.74%
2016-01-01	XLB	3.5%	—	✓ Yes	XLU	4/9	Fed 0.3% YC 1.21%
2016-07-01	XLV	1.8%	—	X No	XLF	7/9	Fed 0.4% YC 0.87%
2017-01-01	XLV	4.3%	—	✓ Yes	XLV	1/9	Fed 0.7% YC 1.25%
2017-07-01	XLE	4.4%	—	✓ Yes	XLK	3/9	Fed 1.1% YC 0.93%
2018-01-01	XLU	8.0%	—	X No	XLY	4/9	Fed 1.4% YC 0.51%

Test date	Top recommendation	P3 score	Actual rel 12mo	Beat SPY?	Actual best	Sector rank	Macro
2018-07-01	XLK	5.1%	—	✗ No	XLV	7/9	Fed 1.9% YC 0.33%
2019-01-01	XLK	18.4%	—	✓ Yes	XLK	1/9	Fed 2.4% YC 0.21%
2019-07-01	XLK	6.7%	—	✓ Yes	XLK	1/9	Fed 2.4% YC 0.25%
2020-01-01	XLU	5.1%	—	✗ No	XLK	6/9	Fed 1.6% YC 0.34%
2020-07-01	XLB	9.4%	—	✓ Yes	XLB	1/9	Fed 0.1% YC 0.52%
2021-01-01	XLV	6.7%	—	✗ No	XLE	6/9	Fed 0.1% YC 0.80%
2021-07-01	XLV	4.9%	—	✗ No	XLK	5/9	Fed 0.1% YC 1.23%
2022-01-01	XLK	8.0%	—	✗ No	XLE	8/9	Fed 0.1% YC 0.79%
2022-07-01	XLP	4.0%	—	✓ Yes	XLE	6/9	Fed 1.7% YC 0.04%
2023-01-01	XLK	5.7%	—	✓ Yes	XLK	1/9	Fed 4.3% YC -0.53%
2023-07-01	XLK	13.1%	—	✓ Yes	XLF	2/9	Fed 5.1% YC -1.06%